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(57) Abstract :
An automated clothing rental kiosk comprising a rectangular frame 101 divided into a first section 102 and a second section 103 connected by a conveyor belt 104, the first section 102 functioning as a clothing storage area and the second section 103 designated for finalizing the selected clothing, a camera 114 for user authentication, a weather determination module for weather-based clothing suggestions, a 3D LiDAR camera 105 for measuring body dimensions, a hanger 106 with a carousel assembly and electromagnetic clips 107 with electromagnetic springs 108, a clothes finalization module including a steaming unit 109, stain removing unit 110, clothes inspection unit, and imaging spectrometer 115 mounted on a linear vertical slider 111 using pneumatic links 112, a Bluetooth/RFID tag on each garment and a reader above the conveyor belt 104, an augmented reality module 116 for displaying lifelike clothing projections, a LED display 113.

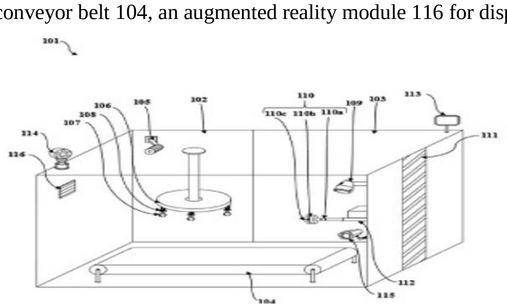

Figure 1

No. of Pages : 22 No. of Claims : 10

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Patentability Search-Analysis Report (Our Ref: MU 1871)

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Analysis Report of Invention

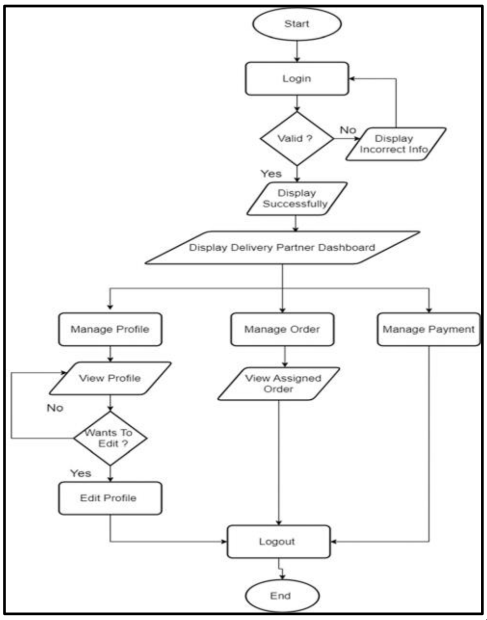
Title: StyleSurfer (rental clothes website)

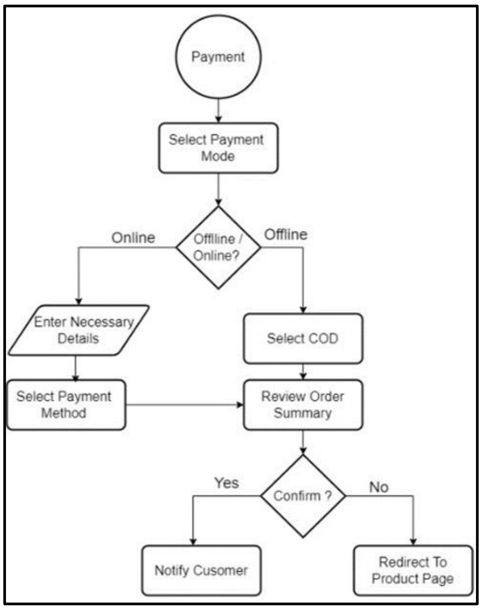
Ref No: MU 1871

Key Features: The present invention is style-on-rent platform that offers trendy, high-quality clothes at affordable prices even for those looking for premium options.

It comprises:

1. Buy Back Option - offers a guaranteed resale value, encouraging eco-conscious buying.
2. Reselling - gives a second life via flash resale, reducing waste.
3. Spin to Win Feature - uses a game-based system to boost user interaction and retention
4. Role-based Dashboards - includes delivery partners with dedicated dashboards
5. Eco-Friendly Vision with Circular Fashion Model - unites all three in one platform, contributing to sustainable fashion practices





Brief Summary of the Search

After performing the patentability search of your invention, the closest found prior art results in the public domain are:

D1.Flyrobe [Link]

D2.Kuro [Link]

D3.JP2016033744 - Rental System And Rental Method [Link]

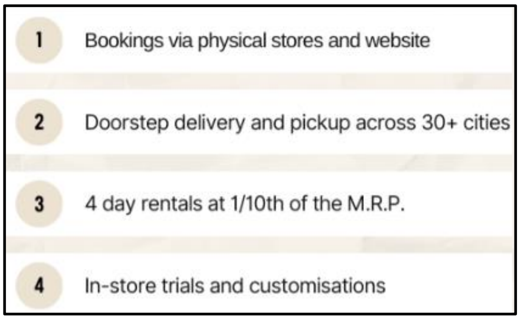
Additional References

D4.IN202311057639A - Fashion Rental System [Link]

D5.KR1020210067165 - Clothing Rental Platform Service System And Method Of Providing Clothing Rental Service Using Same [Link]

Wherein,

D1.Flyrobe **[Link]** – The identified reference discloses **Flyrobe a UNIQUE and PREMIUM RENTAL SERVICE** that is **passionate about FASHION**. It believe that every person deserves to look and feel their best, and we take great joy in helping CUSTOMERS revamp their wardrobes and reinvent their personal STYLE. With an **extensive collection of stylish and TRENDY clothing**, we offer CUSTOMERS the **convenience and affordability of RENTING rather than buying**. In addition to offering convenient and AFFORDABLE FASHION RENTAL services, **Flyrobe is also committed to sustainability**. By choosing to **RENT, CUSTOMERS can enjoy fashionable and TRENDY clothing OPTIONS without contributing to the harmful environmental impacts** of fast FASHION. At Flyrobe, dedicated to **dressing you up and bringing the latest trends to your doorstep**. Let us be your go-to FASHION partner and discover a new way to always look and feel fabulous. This identified reference can be considered in-line w.r.t the subject matter of the invention. However, it does not explicitly mention about spin to win feature.



D2.Kuro **[Link]** – The identified reference discloses **India's 1st 3 in 1 sustainable platform** where one can **rent outfit, shop preloved and sell luxury**. It has access **over 2000 pieces on rotation** with no strings attached. It has **redefine ownership** by renting and reselling. The **pricing is determined through a careful evaluation of the item's**

condition, **brand, purchase year, and overall resale potential**. The **goal is to offer a competitive price that benefits both the seller and buyer.** For International Luxury Brands, **each item undergoes a rigorous two-step verification process**. The in-house team collaborates with **globally recognized authentication partners such as Authenticate First and Real Authentication to ensure every piece is genuine**. For Indian Designer Wear, **items are meticulously inspected by the in-house experts**. This identified reference covers the idea/concept as describes in the given invention. However, it does not explicitly mention about spin to win feature.

D3.JP2016033744 - Rental System And Rental Method [Link] – The identified reference discloses **a rental system** according to the present invention comprises: **business foothold stores**; and a head office system for managing members who are registered. A **user who has a user terminal and is registered as a member can rent the business clothing on the business foothold stores**. In this case, the **registered member can rent business clothes at a clothes laundry business store**. For example, when moving from Tokyo to Osaka, it is possible to move in private clothes, rent business clothes when operating in Osaka, return business clothes in Osaka, and return from Osaka to Tokyo with private clothes. In addition, private clothes may be cleaned in the laundry business store system, and business clothes returned from members can be easily cleaned. The rental system, a **profit point may be provided to a member who provided a questionnaire to the clothing store system by the input system**. In this case, a **profit point can be provided to the member as a compensation for the customer needs of the clothing store**. The profit point **may be a discount ticket, a cash voucher** (can be inferred for gamified engagements) or the like. This identified reference about rental system for business clothes for users once they are registered on the portal having profit point in form of discount and vouchers. However, it does not specifically mention about spin the win feature.

Summary of the Report:

From the identified closest prior results as mentioned in the attached report and their analysis, it has been observed that StyleSurfer (rental clothes website) is **well known** in the prior arts. While doing the exhaustive search,

- Reference D1 & D2 covers all the idea/concept and can be considered in-line with the given invention but both references do not cover the concept of spin to win gamified engagement feature.
- Reference D3 covers rental system for business clothes for users once they are registered on the portal having profit point in form of discount and vouchers.

Based on the references cited above it can be concluded that the features as disclosed in the present invention are already known. The feature of spin to win used for interaction and retention is although not found explicitly in any of the above cited prior arts but this feature along with other features is very obvious for a person skilled in the art hence not adding any novelty to the invention.

In addition, according to section 3(k), “a mathematical or business method or a computer programme per se or algorithms” is not patentable.

Thus, the present invention lacks novelty as well as inventive step.

Therefore, the present invention may not satisfy the patentability criteria.

Status: Seems Non-Patentable

Suggestions/Recommendation: The proposed invention seems not patentable in its present form. Inventors need to work on technological outcomes, as computer programs may be patentable if they demonstrate either a Technical effect or a Technical contribution. Inventions must provide a "technical solution to a technical problem," like enhanced speed, memory optimization, security enhancement, hardware control, etc. It should also be noted that patents encourage innovation and technical advancement, or economic significance over prior art. The concept/ features must fulfil the patentability criteria according to the Indian Patents Act, 1970.

In case, your invention is rejected by Non-Obviousness:

1. Then inventor can prove the comparison report of the present invention and above cited prior art results and provide the information regarding the technical advancement of the present invention wherein comparison of the invention with the cited results can be on the grounds of working, structure, implementation, advantages, etc.

2. Along with the comparison, please provide data for efficiency/efficacy or cost-effectiveness of your invention.

Information: Patentability Criteria for Indian Patent filing:

1. **Novelty:** Invention that is new and unique in the world and there is not any public disclosure about the invention in any previous literature (either patent or Scientific paper or any other literature) basically it should not be known to the public before filing of patent application.

2. **Non-Obviousness Criteria for Patentability (Inventive Step):** Non-obviousness is defined as a sufficient difference from what has been used or described before that a person having ordinary skills in the area of technology related to the invention would not find it obvious to make the change. An invention amounts to be patentable under non-obviousness criteria if several features to be brought together for certain functionality proposed in the invention must not be obvious to a person having ordinary skill in the art. To be more precise, if the features exist alone or in the combination of prior arts, when brought together, a person having ordinary skills in the art must not arrive at the methodology or system of the proposed invention.

3. **Industrial Applicability- "**An invention shall be considered as susceptible of industrial application if it can be made or used in any kind of industry, including agriculture"

4. **Invention and its Subject Matter Should not Fall under Section 3 of Indian Patent act which states about inventions that are not Patentable in India.**

(Link to Section 3)

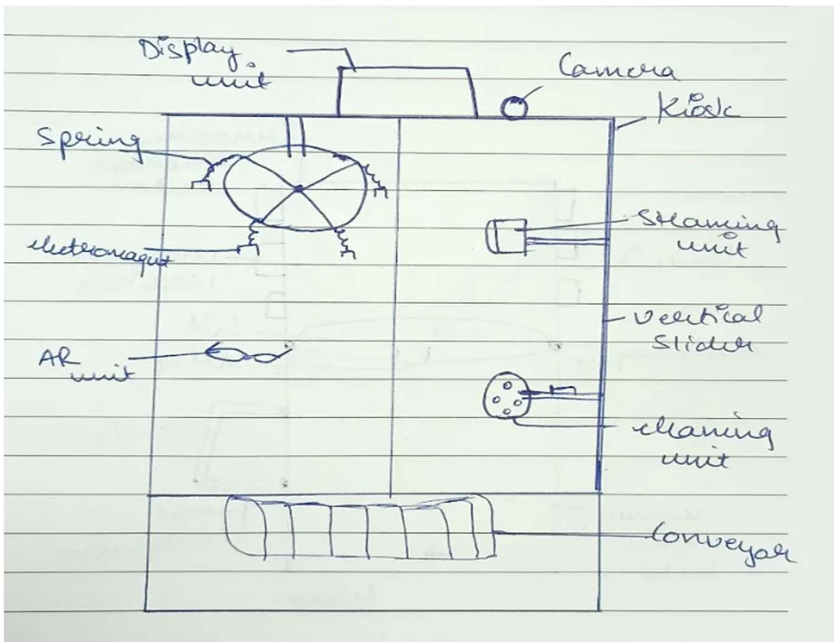
DISCLAIMER

The information, analysis and comments presented herein do not constitute legal advice or opinion on, and are not limited to, patentability, patent infringement, and/or the validity or term of any patent, patent application or claim. Whereas reasonable efforts were made to provide reliable information, the report is not claimed to be exhaustive. It is also possible that additional patents or applications could be interpreted as falling within the scope. Furthermore, new issued patents/applications, modification of patent applications, revocation or abandonment of patents, and evolving case law could result in changes in the analysis contained in this report. Therefore, this report should be seen as an input for illustrative purposes. As with any patent search report, the results are based on available data from public and private databases. Such databases may contain errors or incomplete data. The Patent Analyst makes no warranty as to the integrity of the public and fee-based databases used.

Ref.No.:MU.1871

Title: IoT-Enabled Automated Clothing Rental Kiosk (“StyleSurfer (rental clothes website) “)

Diagram:



Objective:

The device revolutionizes the fashion and rental clothing system by offering personalized, sustainable, and hygienic solutions. Using 3D LiDAR scanning, AI-driven outfit recommendations, and automated cleaning, it ensures perfect fit and pristine garments. Integrated with GPS, temperature sensors, and AR visualization, it suggests weather-appropriate clothing. The mobile app enables real-time tracking, payments, and gamified rentals, while RFID/Bluetooth tags optimize inventory. This scalable system enhances user convenience, reduces returns, and promotes eco-friendly practices in clothing rental shops.

Key Features:

1. The invention discloses a device that transforms clothing rental shops into smart, user-centric hubs, delivering a personalized, sustainable, and hygienic fashion experience.
    - The device uses advanced scanning to capture precise body measurements for perfect fit, while AI-driven recommendations suggest outfits tailored to user preferences and weather conditions.
    - Real-time inventory tracking promotes eco-friendly practices by optimizing stock.
    - An LED display is provided on the device for outfit selection and QR-code payments, this innovative system enhances convenience and redefines sustainable fashion rentals.
2. **The structure of the device consists of a rectangular frame that is divided into two distinct sections, each serving a specific purpose.**
    - The first section functions as a clothing storage area where the user can view and select the type of clothing they wish to use, offering easy access and visibility.
    - The second section is designated for finalizing the selected clothing, where fabric testing, pressing, and stain removal processes are carried out to ensure the clothing is clean, properly pressed, and ready for use.
    - A conveyor is provided in the device connecting both sections for the transferring of clothes as required.
    - **This device will be installed at various tourist locations to facilitate clothing rental services, allowing visitors who have not brought suitable clothing according to the weather conditions they are experiencing while traveling or sightseeing to rent appropriate garments conveniently, thereby ensuring comfort and protection during their visit.**

3. **A temperature sensor is installed on the device, which monitors the ambient temperature in real-time.**
    - **Based on the detection, the device analyzes the current environmental conditions.**
    - **The device then suggests suitable clothing options that are appropriate for the detected temperature. Further, by predicting the weather for the next 3-4 days using an artificial intelligence module, the device suggests suitable clothing options.**
    - **It helps users choose the right clothes to stay comfortable in varying weather conditions.**
4. A GPS (Global Positioning System) is integrated into the device to determine the location where the device is being operated.
    - A weather database integrated with GPS is connected to the device which allows the device to assess the precise weather conditions at the user's current geographic position.
    - The device utilizes this information to suggest appropriate clothing choices based on the observed weather conditions.
    - The device has predictive weather forecasting to forecast conditions over the next few days, enabling users to prepare for anticipated changes in weather and provide options for clothes accordingly.
5. **A 3D LiDAR camera is mounted with the device to measure the body dimensions of the user which ensures a perfect fit for the user.**
    - <u>The 3D LiDAR Camera captures highly accurate body measurements and detailed facial features as soon as the user fed his/her preferences to the device ensuring a seamless user experience.</u>

- The device analyzes this data to extract precise sizing information and evaluates style compatibility based on facial characteristics like skin tone.
- **This process reduces return rates by ensuring perfect fit and enhances personalization by recommending suitable rentals to the user.**

6. **A hanger is installed in the first section of the device arranged with a carousel mechanism for cloth arrangement. This unit has multiple hanging units, each equipped with specialized electromagnetic clips provided with electromagnetic springs that securely hold the cloth inside the device.**
    - **Until the user finalizes the payment for the clothing, the electromagnet will not detach from the clothing, and the electromagnetic spring will ensure that the user can only take the clothes up to a specific distance for trying them on, thereby preventing the clothes from being removed entirely or taken beyond the designated range until the payment process is completed.**
    - When the user inputs their preferences such as style, and color via an LED screen or mobile application, the 3D LiDAR camera precisely analyzes their body measurements and the device cross-references this data with its inventory.
    - The garments corresponding to the user's choice are displayed on the LED screen or mobile application. The user can also upload specific images of the clothes they wish to rent.
    - The user then selects their desired outfit from the presented options, and the carousel mechanism activates, rotating to position the chosen garment for retrieval.

- **When the user finalizes the clothes for rent, a QR code is displayed on the LED screen or the mobile application.**
  - **The user can then directly pay the rental amount using this QR code.**
  - **Once the payment is completed, the clothes are detached from the hanger and fall onto the conveyor belt.**
  - **The conveyor then transfers the clothes for steaming and stain removal, if necessary.**

7. **A linear vertical slider is arranged in the second section of the device where the clothes are finalized. This facilitates the precise vertical movement of components arranged on it.**
   - **A steaming unit is mounted on the linear vertical slider via a pneumatic link.**
     - <u>An imaging spectrometer is arranged inside the section via a pneumatic link.</u>
     - <u>It scans and analyzes the selected dress to detect the presence of any wrinkles or creases.</u>
     - <u>If wrinkles are detected on the garment by the imaging spectrometer, the device automatically triggers the activation of the steaming unit.</u>
     - <u>This steaming process effectively smooths out the wrinkles, ensuring the dress attains a wrinkle-free appearance so that the user can wear it to the occasion.</u>
   - **A stain-removing unit is mounted on the linear vertical slider via a pneumatic link. Multiple nozzles filled with different cleaning solutions are arranged on it.**
     - <u>The imaging spectrometer analyzes the selected dress to detect any stains.</u>

- When a stain is identified, the stain removal unit is automatically activated. The nozzle dispenses a cleaning solution onto the affected area of the fabric.
- A soft brush, attached to the stain removal unit via a ball-and-socket joint, gets activated.
- As soon as the cleaning solution is applied, the brush gently moves, rubbing the stained area to remove the stain.

8. **A fabric inspection unit is mounted on the slider to analyze the fabric quality.**
    - A tactile sensor is integrated with the inspection unit to determine the quality of the fabric.
        - The tactile sensor determines the fabric's softness, flexibility, and tensile strength.
    - A Near-Infrared (NIR) sensor is integrated with the inspection unit to determine the chemical composition of the fabric.
        - NIR sensor analyzes the chemical composition, fiber content, and purity of the fabric.
    - Deep learning algorithms analyze the data from the fabric inspection unit to determine the optimal cleaning solution for the selected garment.
        - Based on this analysis, the device automatically selects the most suitable cleaning solution.
        - The corresponding nozzle is then activated, precisely dispensing the chosen solution onto the fabric to ensure effective stain removal while preserving the material of the dress.
9. An bluetooth/RFID tag is provided on each garment/clothes stored inside the device. A reader is installed above the conveyor belt to read the tag.

- Each tag stores garment data, enabling real-time tracking of the garment.
- The reader reads tag and ensures that users select available outfits, reducing errors in rental shops.
- Tags track rental counts and idle periods, flagging items for reselling.
- This supports sustainability by repurposing over-rented or unused garments.
- The tags log the last cleaning date, ensuring hygiene compliance for each outfit.
- It enhances user trust by providing them with clean and fresh rentals.
- The tags store data such as rental history, condition, etc.
- The device uses deep learning algorithms for predictive analytics for reselling and inventory optimization.

10. **An AR (Augmented Reality) unit is mounted on the device that displays lifelike projections of selected outfits on a reflective panel.**
    - This enhances decision-making by mimicking an in-store try-on experience.
    - The AR unit utilizes data from the 3D LiDAR scans to accurately model the user's body shape and customize the projected garment fit.
    - The projection is adjustable for different viewing angles, ensuring clear visualization from various perspectives.
    - It integrates seamlessly with the mobile app and LED display, providing synchronized visual information and outfit options.
    - It reduces the need for physical try-ons, streamlining the rental process and minimizing contact, thereby promoting hygiene and safety.
11. An LED display is provided on the outer periphery of the device to display real-time data.

- The LED display shows curated outfit recommendations based on user preferences. The user can select from the options presented to them on the LED display.
- **LED display shows real-time rental progress (e.g., "Cleaning in Progress," "Ready for Dispensing").**
- **It keeps users informed about garment preparation, reducing wait time confusion.**
- After the user rents a dress, a QR code is automatically generated by the LED display.
- This allows the users to pay for their rented dresses easily.
- **LED display shows available sizes and stock counts for each recommended outfit.**
- **This feature prevents the selection of unavailable items.**
- **LED display allows users to apply Spin-to-Win coupons, showing potential savings on rentals. It displays coin balance and voucher options, encouraging engagement with the device's gamification.**

12. A mobile application is interlinked with the device with a user-friendly interface.
    - **The user can upload images of clothes of their choice.**
    - **The device then uses image processing to determine the availability of clothes of the user's choice.**
    - The mobile application shows outfit suggestions based on the user's preferences and body measurements.
    - Users can browse and select dresses that fit perfectly from their phone via mobile application.
    - The mobile application lets users spin a wheel to win coins for discounts or free rentals.
    - Users can check their coin balance and redeem vouchers anytime.

- The mobile application shows all past rentals, including sizes and cleaning dates.
- The user can see which dresses the user likes and rent them again easily.
- It helps users plan future rentals.
- The mobile application saves user’s preferences like style, for faster rentals.

13. A microcontroller is installed in the device that serves as the device’s central processing unit, overseeing and synchronizing its components, including the sensors to facilitate seamless operation of the device.

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Marwadi University

INVENTION DISCLOSURE FORM FOR PATENTS

Applicant Name-Marwadi University

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2. **Provide title of the invention: In 100 words or less, please provide an abstract or summary of the invention**

- **Title** :- StyleSurfer

- **Abstract / Summary** :-
StyleSurfer is a style-on-rent platform that offers trendy, high-quality clothes at affordable prices—even for those looking for premium options. It's built using Python and Django for the backend, with HTML, CSS, JavaScript, and Bootstrap on the frontend to create a smooth and easy user experience. Customers can browse and rent outfits with flexible options . Unique features like Buy Back (customize and return or purchase), Reselling (discounted pre-rented outfits), and a Spin-to-Win Coupen discount make the experience exciting. Sellers manage product listings and buy back requests, while delivery partners handle efficient order deliveries via an easy dashboard.

3. **Detail description of the invention:(Answer to all below are required in detail)**

A. Problem the invention is solving:

1. Fixed Rental Days :
   - Problem:
     Most clothing rental websites only allow rentals for specific durations like 3, 5, or 7 days. This doesn't suit everyone, as different events and plans may require clothes for different time periods.
   - Our-Solution:
     With *StyleSurfer*, customers can choose any number of rental days they want based on their needs. Whether it's 2 days for a quick event or 10 days for a long wedding, the flexibility is in their hands.

2. Old or Unsold Stock :-
   - Problem:
     Some clothes are either rented too many times and become worn out, or they are never rented at all. These items pile up in the inventory, leading to waste and losses.
   - Our-Solution:
     We introduced a Reselling feature , where worn or less-rented clothes are sold at a discounted price. This helps clear inventory while offering great deals to customers.

3. No Fun or Rewards :-
   - Problem:
     Many platforms miss out on engaging customers with exciting features or rewards. This makes the experience boring and doesn't encourage people to return often.
   - Our-Solution:
     We've added a Spin to Win feature. Customers can spin a wheel to win coins, which they can collect and redeem for vouchers, discounts, or even free rentals—making the experience fun and rewarding.

4. Customization & Buyback :-
   - Problem:
     If customers want to customize an outfit and later buy it, the process is either too long, confusing, or not available.
   - Our-Solution:
     Our Buy Back feature allows customers to customize outfits (stone, design) and rent them with the option to return later for a pre-decided price. Additionally, if a customer loves the outfit and decides they want to keep it, they can purchase it. This ensures they get something special, and they have the flexibility to either return it or buy it outright.

5. Difficult Dashboards for Sellers & Delivery Partners :-
   - Problem:
     The backend systems on many platforms are not user-friendly. Sellers struggle to manage products, and delivery partners find it hard to handle deliveries efficiently.
   - Our-Solution:
     StyleSurfer provides simple, clean, and easy-to-use dashboards for both sellers and delivery partners. They can easily manage orders, update statuses without any confusion.

B. General Utility/application of the invention:

This invention, called StyleSurfer, is a clothing rental website that solves all these issues. It gives more control and freedom to both customers and sellers.

Key uses:

- Rent clothes for any number of days you want.
- Buy used products at discounted prices if they are not getting rented anymore.
- Play a spin-the-wheel game to win coins and use them to get discount coupons.

- Customize clothes and get them made as per your choice.
- Buy back customized products before returning them.
- See products in multiple sizes and choose easily.
- Simple dashboards for sellers and delivery partners to manage everything easily.

C. Advantages of the invention disclosing about the increased efficiency/efficacy :

1. Rent Your Way – Instead of being tied to fixed rental periods, users have the freedom to choose exactly how long they want to keep an item. This brings more flexibility to suit different occasions and plans.

2. Smart Resell Feature – Products that are either well-rented or have been idle for a while are moved to a special resale zone with attractive discounts. This ensures sellers can clear stock efficiently while still earning from older items.

3. Fun with Spin & Win – To add a playful twist to the experience, users get to spin a lucky wheel twice a month. They can win coins which are redeemable for coupons, making savings feel like a reward.

4. Personalized Product Buy-Back – Customers can request personalized touches on select products by submitting a customization form. Once approved, the product is created, rented, and later, the customer even has the option to purchase it back.

5. Smarter Size Tracking – Every product comes with detailed size availability. Each size has its own inventory count, so users can easily check which sizes are ready to rent or buy.

6. Intuitive Dashboards for Sellers & Delivery Partners – Dedicated, user-friendly panels allow sellers and delivery agents to stay on top of everything – from product listings and orders to deliveries – all in one place, with minimal hassle.

D. Best way of using the invention as well as possible variants :

- Best Way to Use:
  - A customer browses the website, picks a product, and sets their own rental duration based on how long they actually need it – no fixed days required.
  - They get access to a monthly spin game, where they can try their luck up to two times to earn coins and unlock exclusive discounts.

- When a product has been rented multiple times or is not receiving enough rental requests, the team or seller can manually move it to the resale section. This gives the product a second life by offering it at a discounted price, allowing customers to purchase it at a lower cost instead of just renting.

- If someone wants a personalized outfit, they simply fill out a customization form. Once the seller reviews and approves it, the item is crafted, rented out, and later the customer gets a chance to buy it before the return period ends.

- Every product page displays all the available sizes along with how many pieces are in stock, helping customers make quick and clear decisions.

- Behind the scenes, sellers and delivery partners manage their tasks easily through a dashboard that shows all orders, requests, and updates in a user-friendly format.

- Possible Variants and Add-ons:
  - AI-powered outfit suggestions based on each user's style, preferences, and browsing habits.
  - Virtual try-on feature so users can preview how clothes will look on them before making a rental decision.
  - 360° product view, allowing users to see the clothing item from every angle for better understanding of fit, fabric, and detailing.
  - Location-based pricing and delivery options, offering faster service and flexible costs depending on the user's area.

E. Working of invention along with Drawing, schematics and flow diagrams if required with complete explanations:

1. Buy Back Flow :-
   - Customer Side:
     - Customer Logs In
     - Goes to 'Buy Back' Section from Navbar
     - Browse Available Products Eligible for Buy Back
     - Clicks on a Product → Views Full Details
     - Selects Customization Options (Fabric, Stones, Design, etc.)
     - Uploads a Reference Image or Design Sketch

- Clicks on ‘Submit Design’

- Seller Side:
  - Seller Receives Customization Request
  - Reviews Design + Image + Customization Preferences
  - Accepts or Declines the Request

- Back to Customer Side:
  - Customer Gets Status Update in 'My Account' > 'Buy Back Requests'
  - If Approved → ‘Buy Back Now’ Button Appears
  - Customer Selects Expected Delivery Date
  - Clicks on ‘Buy Back’ → Product Added to Cart
  - Proceeds to Payment and Places Order

- Post-Order (Before Return Period):
  - Product Delivered to Customer
  - Before 5 Days of Return Window Expiry → A Button Appears:
    - [Purchase Permanently]
  - If Clicked → Pays Remaining Amount (if any), Product is Theirs
  - If Not → Default Process is Return Pickup
  - If Returning → Seller Reviews Condition
  - If Product Meets Condition → Buy Back Price is Refunded

2. Reselling :-
   - Seller Side:
     - Seller Logs In
     - Goes to ‘My Products’ Section
     - Filters Products by Category
     - Sees List of Owned Products (with Rental Count Info)
     - For Each Product → ‘Reselling’ Button is Available
     - Clicks on ‘Reselling’ Button
     - Popup/Alert Appears: “Are you sure you want to resell this product?”
     - Seller Clicks ‘OK’
     - Product Status is Updated to ‘Reselling’
     - Now Visible in Customer’s ‘Reselling Section

- Customer Side:
  - Customer Logs In
  - Goes to ‘Reselling’ Section
  - Browses Reselling Products
  - Clicks on a Product → Opens Product Description Page
  - Clicks ‘Add to Cart’
  - Goes to Cart → Proceeds to Checkout
  - Places Order and Makes Payment
  - Product Delivered → Ownership Transferred → Not Available for Rent

3. SpinToWin :-
   - Customer Side:
     - Customer Logs In
     - Goes to ‘Spin to Win’ from Navbar
     - Spin Button is Enabled (Limit: Only 2 Spins Per Month) → (If Limit Reached: Show Messag“Spin Limit Reached for This Month”)
     - User Clicks on ‘Spin’
     - Wheel Spins and Lands on a Reward (e.g., 10 Coins, 20 Coins, 50 Coins)
     - Popup Message Appears: “You Won X Coins!”
     - Coins are Added to User’s Wallet
     - User Can View Coin Balance in:
     - Account Section (My Coins)
     - Spin to Win Section
     - Customer Clicks on ‘Redeem Vouchers’ in Spin to Win Section
     - All Available Vouchers Are Displayed with Required Coin Values
     - Customer Selects a Voucher → Clicks ‘Redeem’
     - Voucher Code is Generated and Stored in Account Section (‘My Vouchers’)
     - Customer Adds Product to Cart
     - During Checkout → Applies Voucher Code from Account Section
     - Discount Applied → Proceeds to Payment and Completes Order

4. Have you conducted Primary Patent Search? Yes / No (if yes, attach the patent search results)

- No

5. **List out the known ways about how others have tried to solve the same or similar problems? Indicate the disadvantages of these approaches. In addition, please identify any prior art documentation or other material that explains or provides examples of such prior art efforts**

S. No.	Existing state of art	Drawbacks in existing state of art	Overcome (how your invention is overcoming the drawback)
1	Predefined Rental Durations Platforms like Rent the Runway and Flyrobe offer rentals for set periods (e.g., 3, 5, or 7 days).	- Lack of flexibility for customers needing different rental durations. - May not align with individual event schedules.	- Allows customers to select any rental duration based on their specific needs, enhancing user satisfaction and flexibility.
2	Resale of Pre-worn Items Companies like Rent the Runway sell pre-worn items after multiple rentals.	- Items are automatically moved to resale after a set number of rentals, regardless of current demand. - Limited focus on traditional Indian attire in resale offerings.	- Empowers sellers to manually decide when to move items to resale, considering demand and condition, optimizing inventory management and profitability. - Includes traditional Indian clothing in resale options, catering to a broader audience.
3	Basic Incentive Features Some platforms offer minimal reward systems for user engagement.	- Incentives often limited to points or badges without substantial benefits. - May not significantly enhance user engagement or retention.	- Introduces a 'Spin to Win' feature where users can earn coins redeemable for discounts, providing tangible benefits and increasing user engagement.

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4	Manual Customization and Buyback Processes Limited platforms offer customization with buyback guarantees.	- Customization processes are often manual and time-consuming. - Lack of streamlined communication between customers and sellers.	- Provides an integrated system where customers can request customizations, receive approvals, and have the option to purchase the item post-rental, streamlining the entire process.
5	Complex Seller and Delivery Dashboards Existing platforms offer dashboards for sellers and delivery partners.	Dashboards can be complex and not user-friendly. - Difficulty in managing orders and inventory efficiently.	- Offers intuitive and easy-to-navigate dashboards for sellers and delivery partners, facilitating efficient order management and tracking.

6. List the Technical features and Elements of the invention along with the Description of your invention from start to end.:

- **Technical Features:**
  - User-Friendly Interface: StyleSurfer uses HTML, CSS, JavaScript, and Bootstrap to provide an attractive and easy-to-use interface.
    - Python Django Framework: Django is used for the backend to ensure a secure and organized structure with proper database integration.
    - User Roles Integration: The system supports different types of users with role-specific functionalities—Customer, Seller, Admin, and Delivery Partner.
    - Secure Payment Integration: Features are developed for integrating payment gateways like stripe for smooth and secure transactions.
    - Custom Features: Innovative functionalities like reselling , Buy Back Option, and Spin to Win are added to make the platform stand out.
    - Responsive Design: The website is responsive and works smoothly across devices like mobiles and tablets.

- **Elements of the Invention:**
  - Modules:
    - Customer Module

- Seller Module
- Admin Module
- Delivery Partner Module

- Components:
  - Product Listing
  - Cart and Checkout
  - Rental and Purchase Options
  - Reselling and Buy Back
  - Review & Rating System
  - Spin to Win Game Feature

- Databases:
  - Uses SQLite3 to manage data related to users, products, orders

- Interfaces:
  - Customer-Facing Frontend
  - Seller Dashboard
  - Dp dashboard
  - Admin Panel

- **Description from Start to End:**
  - Beginning: The idea for StyleSurfer started with a simple thought providing people a better alternative to buying expensive clothes for weddings and parties, which are usually worn once and then left unused. In cities like Rajkot, many people spend a lot on such clothing and never use it again. Our aim was to reduce this kind of waste by creating a rental platform.

  - Idea and Research: We realized there should be a service where people can rent clothes instead of buying them, promoting sustainability and making fashion more accessible. The focus was on stylish, trendy, and reusable fashion.

  - Design and Development Steps:
    - Planned the structure of the website.
    - Started backend development using Django Framework.
    - Developed modules separately – Customer, Seller, Admin, Delivery Partner.
    - Added special features like Spin to Win, Buy Back, and Reselling.
    - Designed a responsive UI using HTML, CSS, and Bootstrap.

  - Created models and database structure to support Product Management and Rental System.

- Functioning and Usage:
  - Customers can register/login and rent clothes.
  - Sellers can list their garments on the platform.
  - Admins manage the overall system and users.
  - Delivery Partners handle order deliveries and returns.
  - Customers can win rewards using the "Spin to Win" feature.
  - With the "Buy Back" feature, customers can purchase clothes and return them later for a guaranteed refund amount.

- Final Outcome: StyleSurfer is a platform that makes fashion more affordable, accessible, and eco-friendly. Clothes that have been rented several times can be sold under the "Reselling." Features like "Buy Back" provide value-for-money options, helping both customers and the environment.

7. **List out the features of your invention which are believed to be new and distinguish them over the closest technology.**

- Buy Back Option (New & Unique)
  - What it is: Customers can buy customized clothes and return them later for a guaranteed fixed amount.
  - Difference: Most rental platforms allow only rent or purchase. StyleSurfer offers a guaranteed resale value, encouraging eco-conscious buying.

- Reselling (Innovative Selling Model)
  - What it is: Clothes that have been rented multiple times are sold at discounted prices under limited-time deals.
  - Difference: Traditional rental sites discard or store such items. StyleSurfer gives them a second life via flash resale, reducing waste.

- Spin to Win Feature (Gamified Engagement)
  - What it is: Users spin a wheel to win coins or rewards that can be redeemed for discounts or vouchers.
  - Difference: Unlike other platforms that only offer referral points or seasonal sales, StyleSurfer uses a game-based system to boost user interaction and retention.

- Role-based Dashboards (User-Specific Experiences)
  - What it is: Separate and well-organized interfaces for Customer, Seller, Admin, and Delivery Partner.

- Difference: Other platforms often focus only on customers and sellers. StyleSurfer includes delivery partners with dedicated dashboards.

- Eco-Friendly Vision with Circular Fashion Model
  - What it is: Combines rental, resale, and buy-back to promote sustainable fashion.
  - Difference: While most platforms focus on one service (either rental or resale), StyleSurfer unites all three in one platform, contributing to sustainable fashion practices.

8. Has the invention been built or tested or implemented? If yes please provide the Efficiency/Efficacy details of the invention

- Testing/Implementation and Efficiency/Efficacy of the Invention:

  - Development and Status:
    - StyleSurfer, the fashion rental website, is currently a working prototype. It has been built, but it hasn't been tested or launched yet. The backend of the website was developed using Python and Django, and it includes basic features like user registration, managing products, and processing orders. The frontend is designed using HTML, CSS, JavaScript, and Bootstrap to create the website's look and feel. While the core features are in place, the website has not yet been tested or made available for use by the public.

  - Efficiency:
    - At this stage, we haven't fully tested how efficient StyleSurfer is in real-world situations. Based on the development so far, we expect the system to perform well in terms of server response time, how it uses resources, and its ability to handle growth. However, these expectations haven't been tested in a live environment with many users or high traffic. More testing will be needed to confirm how well the site performs when it's actually up and running.

  - Efficacy:
    - The efficacy of StyleSurfer, including its core functionalities such as product browsing, rental processing, buy-back feature, and special promotions (e.g., spin-to-win), has not been formally tested. The intended efficacy is based on the design goals of improving user experience, ensuring seamless transactions, and fostering engagement with the rental system. Testing will be required to confirm its performance against these objectives and to evaluate how effectively it meets user needs.

9. **Briefly state when and how you first conceived this idea?**

- I came up with the idea for StyleSurfer during my 5th semester while exploring project ideas that combine fashion, technology, and sustainability. While researching fashion rental websites, I noticed that most platforms only focus on renting and returning clothes — they lacked fun, flexibility, and eco-friendly ideas. That’s when I decided to build something more engaging and useful.

- Here are the key features I added and why:

    - Flexible Rental Duration
        - Customers can choose how many days they want to rent an outfit whether it's 2 days for a party or 10 days for a wedding. This gives them full control and better convenience compared to fixed rental periods.

    - Buy Back
        - Many people want to wear custom-designed or expensive outfits just once but avoid them due to the high cost. With this feature, users can customize an outfit and return it later for a guaranteed buy-back price. It saves money and promotes reuse.

    - Reselling
        - Once an outfit has been rented multiple times, it may not be in demand for rentals anymore. Instead of letting it go to waste, sellers can resell it at a discounted price. This reduces waste and gives customers access to affordable fashion.

    - Spin to Win
        - To make the platform more interactive, I added a fun reward system where users can spin a wheel to win coins. These coins can be used to redeem vouchers or discounts, making the experience more exciting and engaging.

- All of these features were implemented using Python Django, making StyleSurfer not just a fashion rental website but a smart, sustainable, and enjoyable platform for users.

10. **Have you sold, offered for sale, publicly used or published anything related to this invention? If yes, please briefly explain the dates and circumstances. List those individuals to whom you have revealed your invention. Were non-discloser documents signed prior to discloser in each case? Please state any deadlines of which you may be aware for filing an application on this invention.**

    - No

11. **Include any reasons that your invention would not have been obvious to someone of average skill in the art.**

    - Our Buy Back feature is a new idea that people wouldn't easily think of. While there are rental services and resale options out there, none of them let you customize the clothes you rent and then return them for a guaranteed price. Plus, if you like the customized clothes, you can even buy them for yourself.

    - Most rental services just focus on renting clothes or selling used clothes. They don't let you change the clothes while renting or promise a buy-back price. A "buy-back price" is a guaranteed amount of money that the customer can get back if they decide to return the item after renting it, even if it has been customized. This is different from other services that don't offer this kind of option. Our idea combines both customization and a clear plan for how you can return the clothes for a set price. This gives customers something new and special.

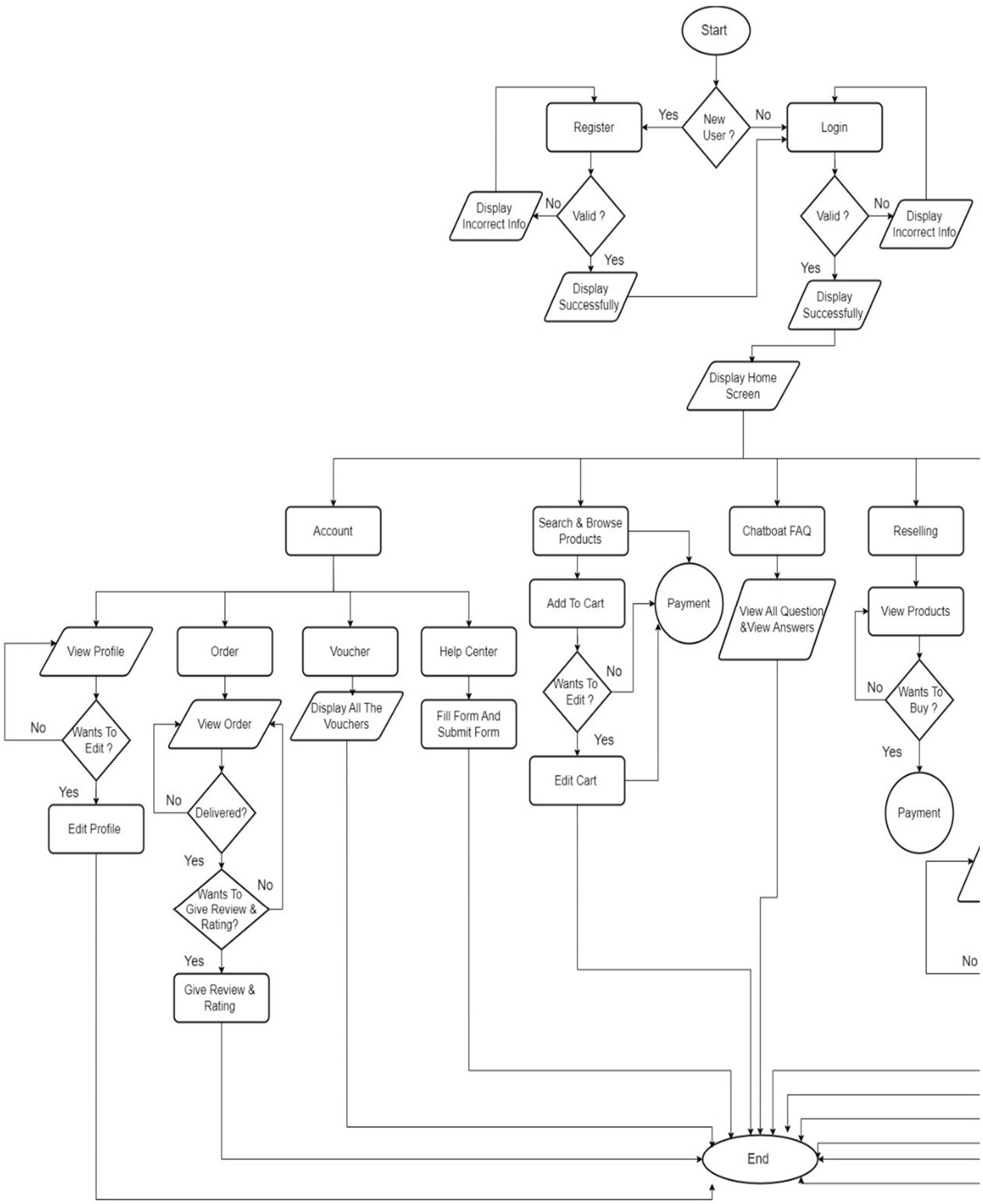
    - The reason why others wouldn't have thought of this is because combining customization with a buy-back option is quite complex. Rental services usually deal with standard clothing items, so adding customization and setting a fair buy-back price based on changes to the item and its rental history requires more planning and flexibility. It's not something the industry commonly focuses on, which is why this idea is unique and not obvious.

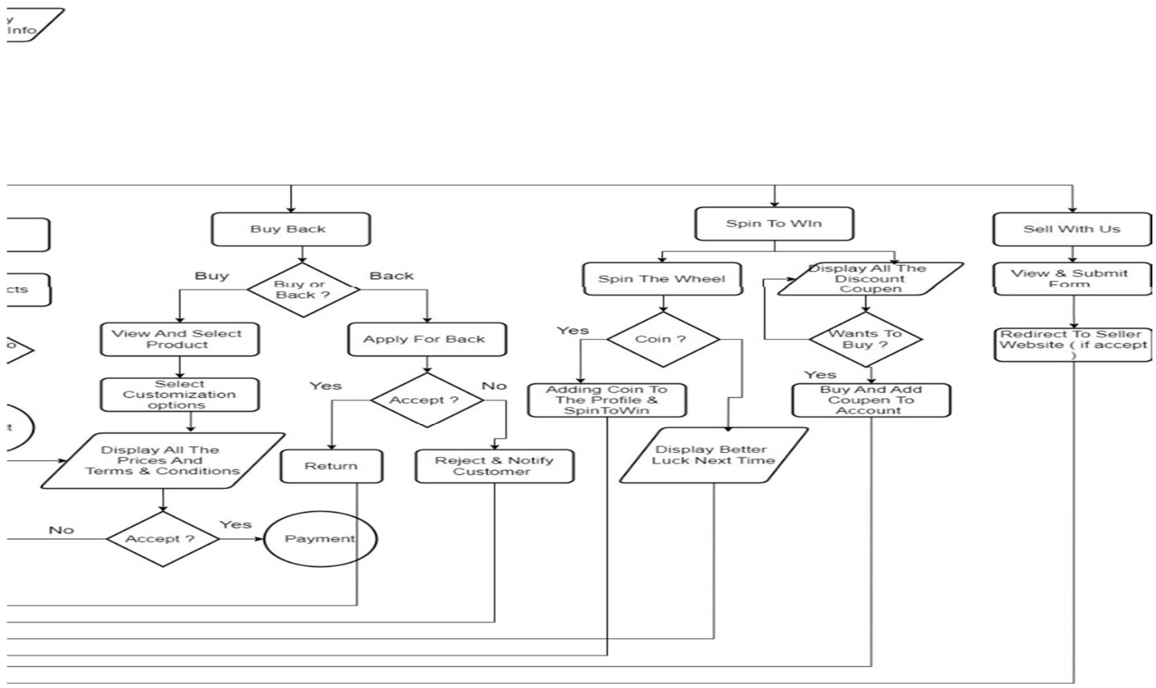
12. **Additional comments by inventor (if you want to give more details out of scope of this IDF).**

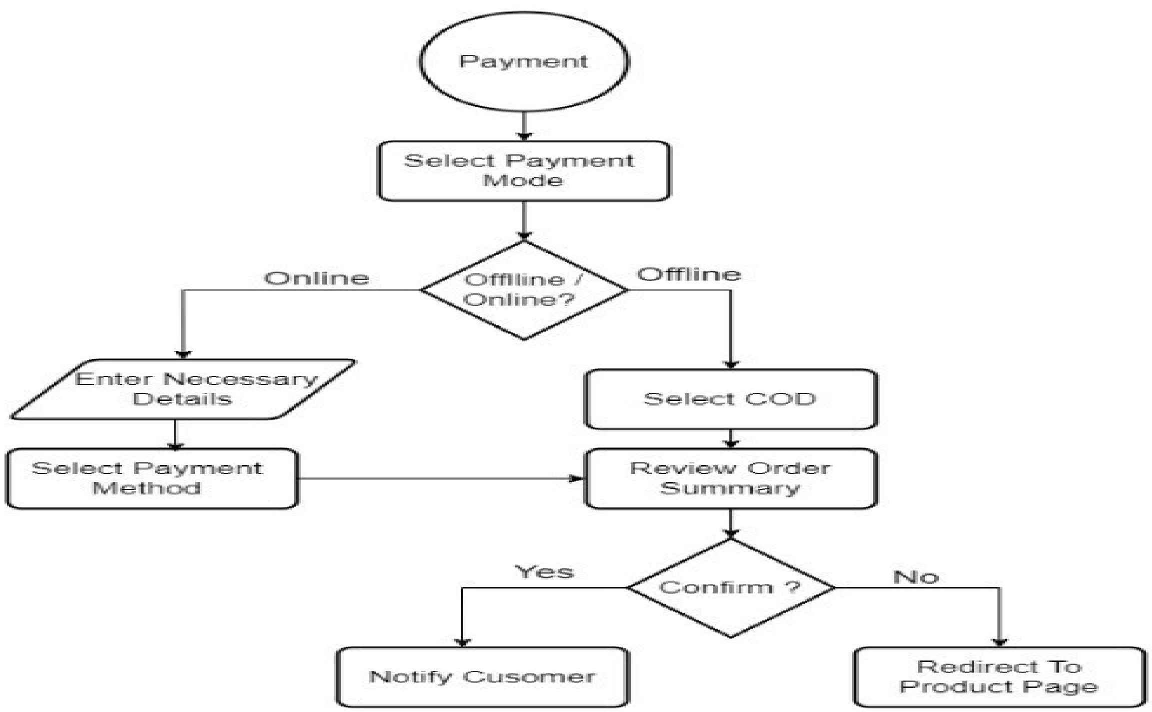
    - As a students working on this project, we would also like to mention some additional ideas that could improve the user experience even further. In the future, this invention could include Augmented Reality (AR) technology, which would allow users to virtually try on clothes or view the product in real-time using their mobile device or computer camera. Another idea is to add a 360-degree view feature, so users can rotate and view the clothing item from every angle. These features are not part of the current version but can be added later as improvements or considered for a separate patent in the future.

13. Drawings/Flowchart/Table :-

- Customer Flowchart :




Info
Buy Back
Spin To Win
Sell With Us
Buy
Buy or Back ?
Back
Spin The Wheel
Display All The Discount Coupen
View & Submit Form
View And Select Product
Apply For Back
Yes
Coin ?
Wants To Buy ?
Redirect To Seller Website (if accept)
Select Customization options
Yes
Accept ?
No
Adding Coin To The Profile & SpinToWin
Yes
Buy And Add Coupen To Account
Display All The Prices And Terms & Conditions
Return
Reject & Notify Customer
Display Better Luck Next Time
No
Accept ?
Yes
Payment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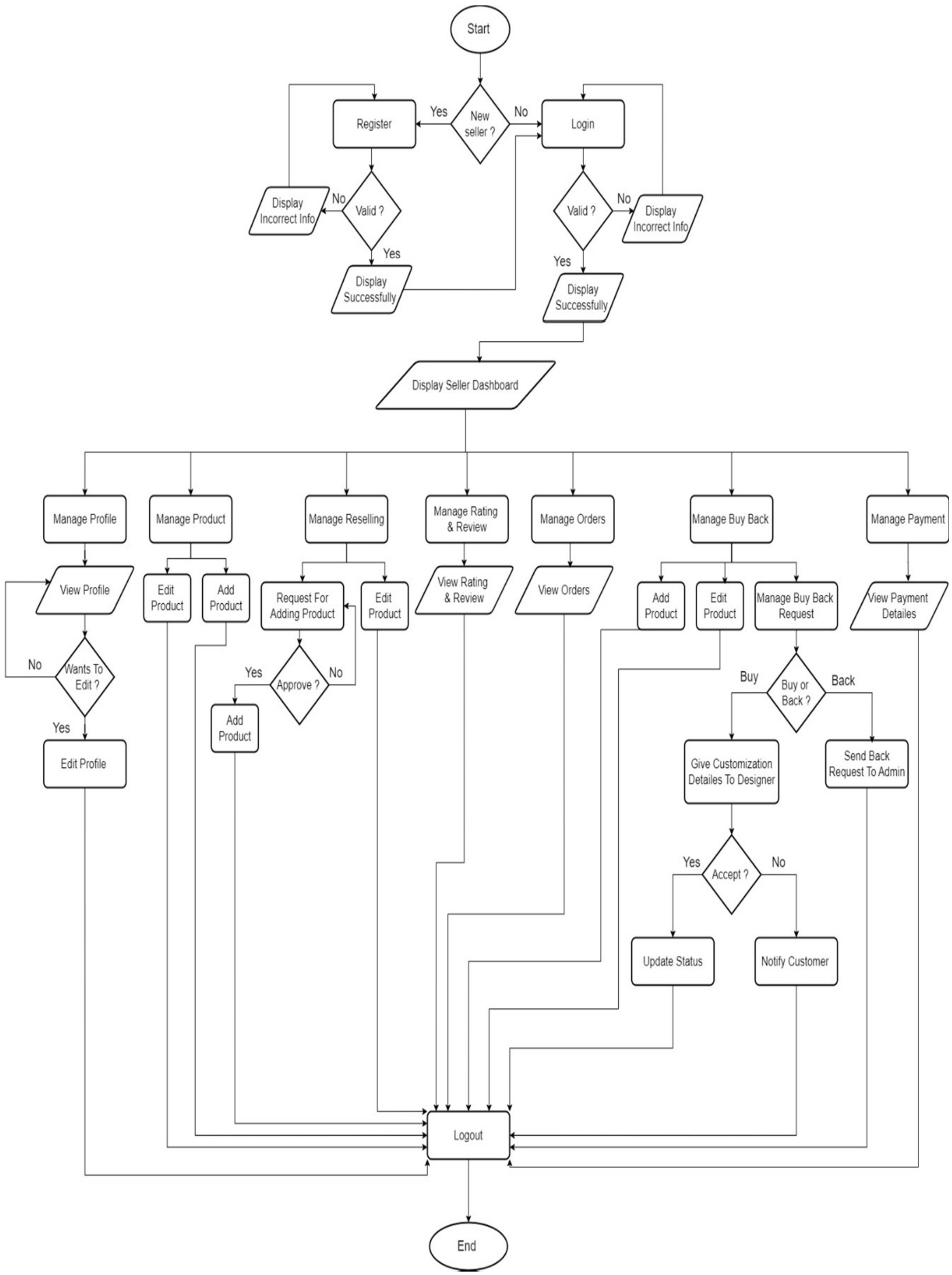
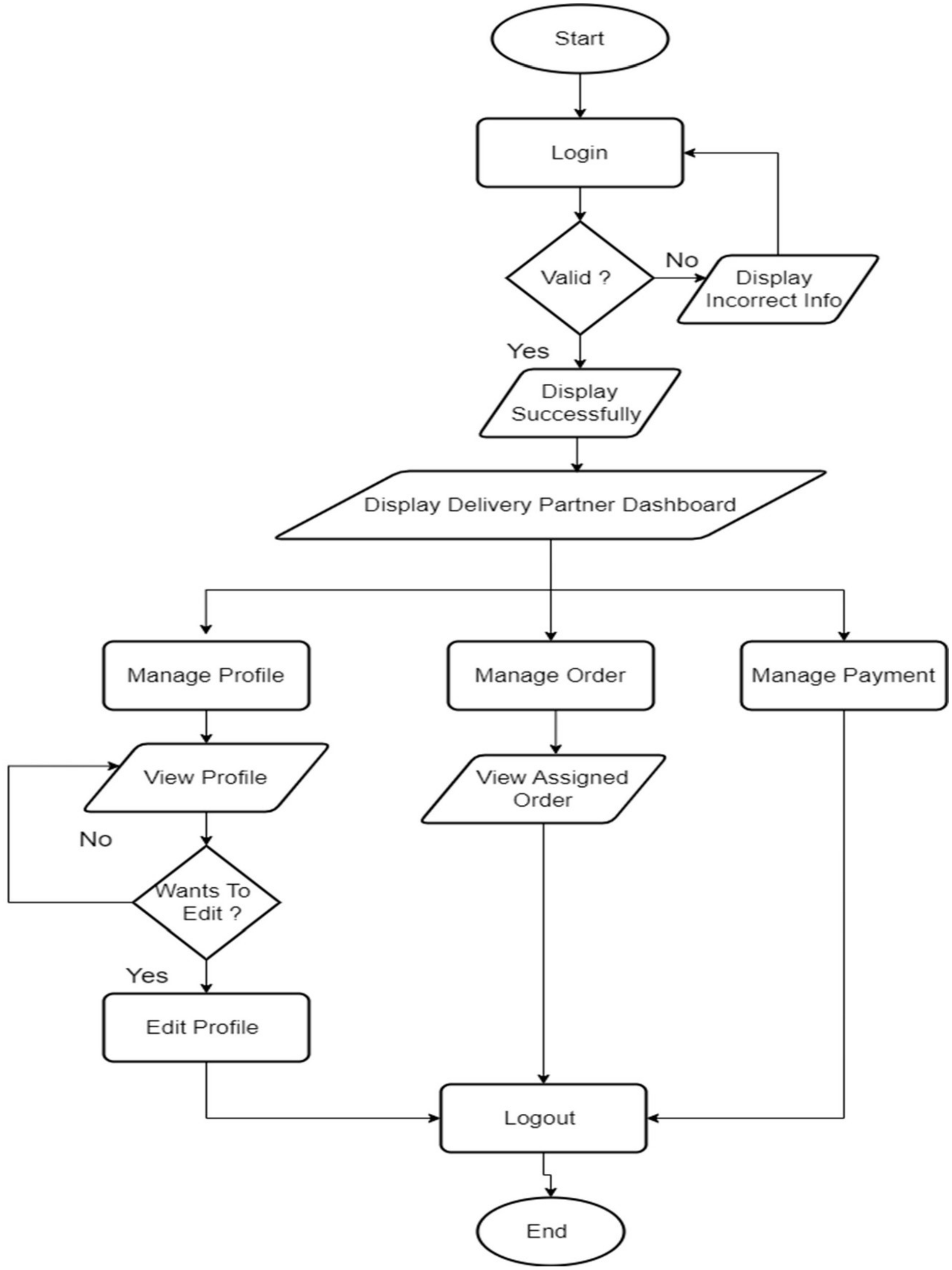
Select Payment Mode
Online
Offline / Online?
Offline
Enter Necessary Details
Select COD
Select Payment Method
Review Order Summary
Yes
Confirm ?
No
Notify Cusomer
Redirect To Product Page

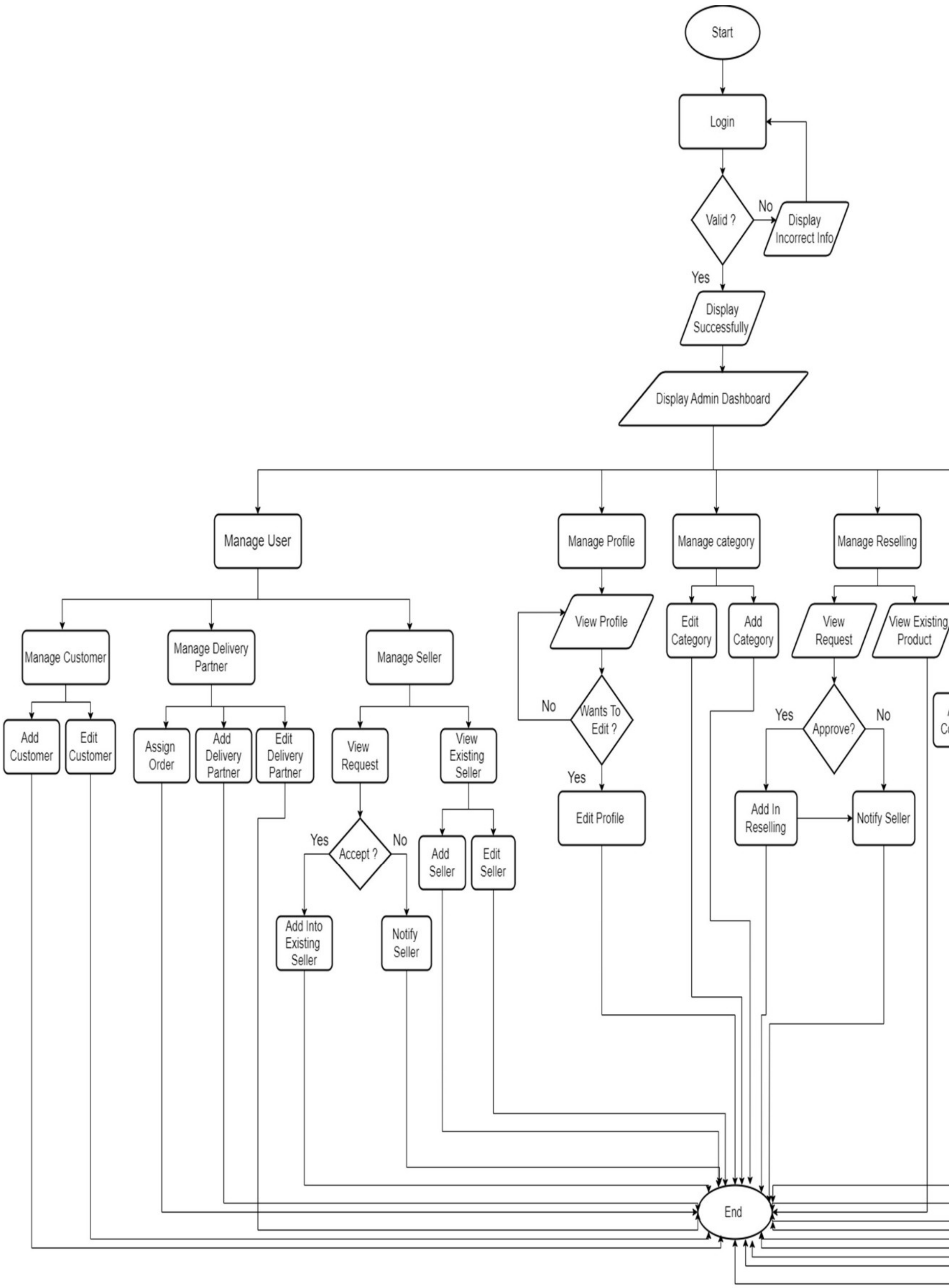
- Seller Flowchart :

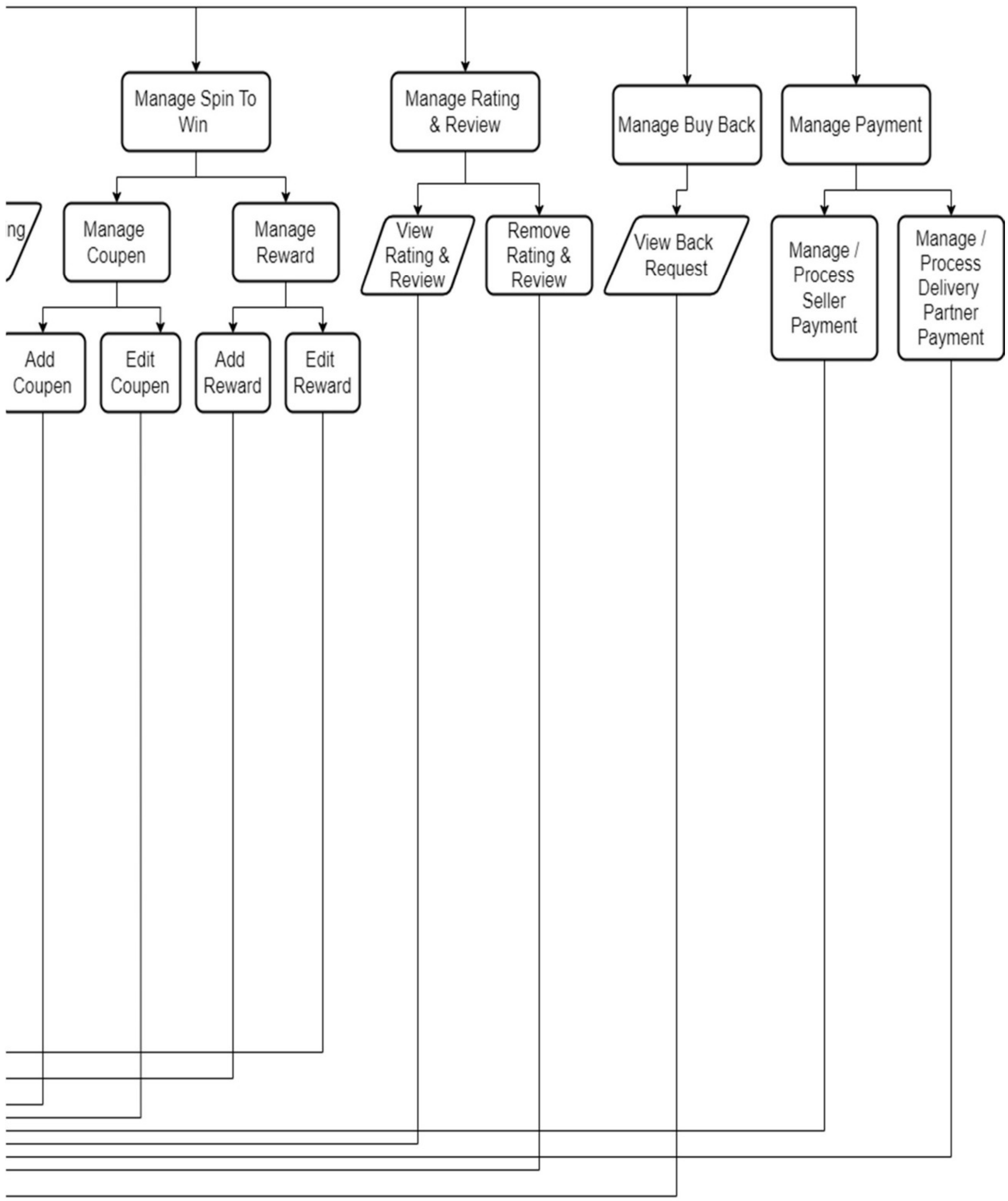


- Delivery Partner Flowchart :



- Admin Flowchart :




Manage Spin To Win
Manage Rating & Review
Manage Buy Back
Manage Payment
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Manage Coupen
Manage Reward
View Rating & Review
Remove Rating & Review
View Back Request
Manage / Process Seller Payment
Manage / Process Delivery Partner Payment
Add Coupen
Edit Coupen
Add Reward
Edit Reward